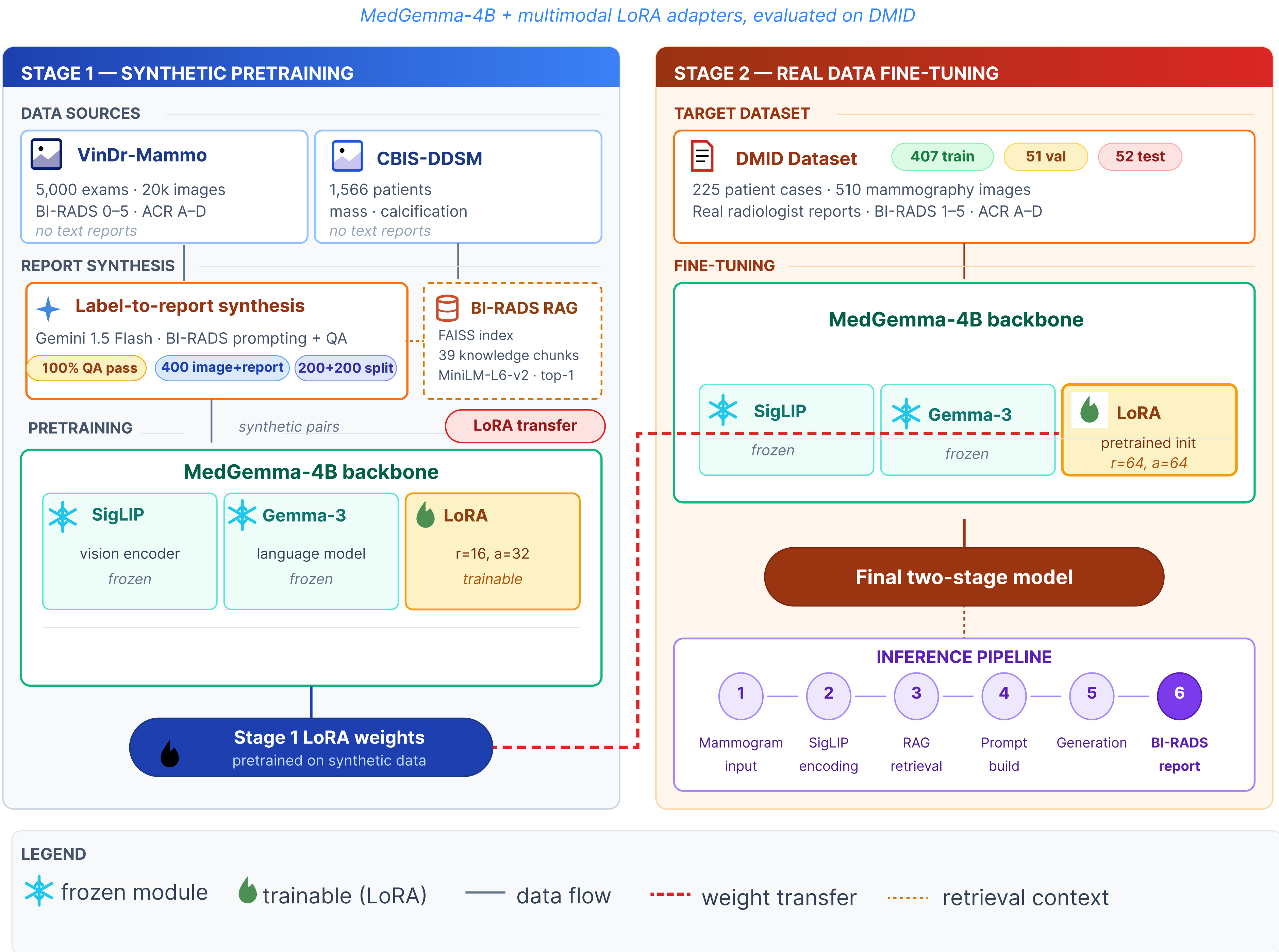
MedGemma-4B + multimodal LoRA adapters, evaluated on DMID
STAGE 1 — SYNTHETIC PRETRAINING
DATA SOURCES
VinDr-Mammo
5,000 exams · 20k images
BI-RADS 0–5 · ACR A–D
no text reports
CBIS-DDSM
1,566 patients
mass · calcification
no text reports
REPORT SYNTHESIS
Label-to-report synthesis
Gemini 1.5 Flash · BI-RADS prompting + QA
100% QA pass
400 image+report
200+200 split
BI-RADS RAG
FAISS index
39 knowledge chunks
MiniLM-L6-v2 · top-1
PRETRAINING
synthetic pairs
LoRA transfer
MedGemma-4B backbone
SigLIP
vision encoder
frozen
Gemma-3
language model
frozen
LoRA
r=16, a=32
trainable
Stage 1 LoRA weights
pretrained on synthetic data
STAGE 2 — REAL DATA FINE-TUNING
TARGET DATASET
DMID Dataset
407 train
51 val
52 test
225 patient cases · 510 mammography images
Real radiologist reports · BI-RADS 1–5 · ACR A–D
FINE-TUNING
MedGemma-4B backbone
SigLIP
frozen
Gemma-3
frozen
LoRA
pretrained init
r=64, a=64
Final two-stage model
INFERENCE PIPELINE
1 Mammogram input
2 SigLIP encoding
3 RAG retrieval
4 Prompt build
5 Generation
6 BI-RADS report
LEGEND
frozen module
trainable (LoRA)
data flow
weight transfer
retrieval context